

Bundle Structures in Topology

Labix

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Abstract

- Notes on Algebraic Topology by Oscar Randal-Williams

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1 Vector Bundles

1.1 Basic Definitions

Definition 1.1.1 (Vector Bundles)

Let F be a field. Let E, B be spaces with B connected. Let $p : E \rightarrow B$ be a continuous map. We say that p is a vector bundle of dimension $d \in \mathbb{N}$ over F if the following are true.

- $p : E \rightarrow B$ is surjective.
- Local Triviality: There exists a collection of homeomorphisms

$$\{(U_i \subseteq B, \phi_i : p^{-1}(U_i) \rightarrow U_i \times F^k) \mid i \in I\}$$

called local trivialization charts such that $B = \bigcup_{i \in I} U_i$ and the following diagram commutes:

$$\begin{array}{ccc} p^{-1}(U_i) & \xrightarrow{\phi_i} & U_i \times F^k \\ & \searrow p \quad \swarrow \pi & \\ & U_i & \end{array}$$

where π is the projection by forgetting the second variable.

We say that B is the base space, E the total space. It is denoted as (F, E, B) .

The local trivialization means that locally at a neighbourhood, the vector bundle looks the same the open set times F^k . In particular, there is also a notion of trivial bundle which means that the bundle is globally just $B \times \mathbb{R}^r$.

Definition 1.1.2 (The Fiber of a Bundle)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let $b \in B$. Define the fiber of E at B to be

$$E_b = p^{-1}(b)$$

Example 1.1.3 (The Trivial Bundle)

Let B be a space. Let F be a field. The trivial bundle $p : B \times F^k \rightarrow B$ defined by $p(b, v) = b$ is a vector bundle. The bundle is denoted by e^k .

Example 1.1.4 (The Mobius Strip)

Let $E = \frac{I \times \mathbb{R}}{\sim}$ where $(0, x) \sim (1, -x)$ for $x \in \mathbb{R}$. Then the projection map $E \rightarrow \frac{I}{0 \sim 1} \cong S^1$ is a vector bundle of rank 1.

Lemma 1.1.5

Let F be a field. Let $p : E \rightarrow B$ be a vector bundle of dimension $d \in \mathbb{N}$ over F . Then the local trivialization charts induces a homeomorphism

$$E_b \cong F^d$$

for all $b \in B$.

Definition 1.1.6 (Morphism of Vector Bundles)

Let $p_1 : E_1 \rightarrow B_1$ and $p_2 : E_2 \rightarrow B_2$ be vector bundles. A morphism of these vector bundles is given by a pair of continuous maps $f : E_1 \rightarrow E_2$ and $g : B_1 \rightarrow B_2$ such that the following diagram commutes

$$\begin{array}{ccc} E_1 & \xrightarrow{f} & E_2 \\ \downarrow p_1 & & \downarrow p_2 \\ B_1 & \xrightarrow{g} & B_2 \end{array}$$

and $f|_{p^{-1}(b)}$ is a linear map for all $b \in B$.

Definition 1.1.7 (Isomorphism of Vector Bundles) A bundle homomorphism from E_1 to E_2 is an isomorphism if there exists an inverse bundle homomorphism from E_2 to E_1 . In this case, we say that E_1 and E_2 are isomorphic.

Lemma 1.1.8

Let B be space. Let $p_1 : E_1 \rightarrow B$ and $p_2 : E_2 \rightarrow B$ be vector bundles. Let $f : E_1 \rightarrow E_2$ be a morphism of vector bundles. Then f is an isomorphism if and only if $f|_{p_1^{-1}(b)}$ is an isomorphism for all $b \in B$.

Proof

If f is an isomorphism, then the inverse of $f|_{p_1^{-1}(b)}$ is clearly given by $f^{-1}|_{p_2^{-1}(b)}$.

Now suppose that $f|_{p_1^{-1}(b)}$ is an isomorphism for all $b \in B$. Then clearly f is bijective. I claim that $p_2 = p_1 \circ f^{-1}$. Let $x \in E_2$. Then there exists $y \in E_1$ such that $f(y) = x$. Then $p_1(f^{-1}(x)) = p_1(y)$. But $p_1 = p_2 \circ f$ by definition of morphisms of vector bundles. Hence $p_1(y) = p_2(f(y)) = p_2(x)$ and we conclude.

It remains to show that f^{-1} is continuous. Let $U \subseteq E_1$ be an open set. Then there exists local trivialization charts (V_i, ϕ_i) of E_1 such that the V_i 's cover $p_1(U)$. Then we have $U = \bigcup_{i \in I} U \cap V_i$ and $f^{-1}(U)$ is open if and only if $f^{-1}(U \cap V_i)$ is open. But since restrictions of local trivialization charts are also local trivialization charts, it suffices to prove that f is continuous on local trivialization charts.

So let (U, ϕ) be a local trivialization chart of E_1 . Since E_2 is also covered by local trivialization charts, WLOG assume that (U, ψ) is a local trivialization chart of E_2 . Now the map $\psi \circ f \circ \phi^{-1} : U \times F^k \rightarrow U \times F^n$ sends (x, v) to $(x, g(x)(v))$ for some function $g(x) : F^k \rightarrow F^n$. Since the map is a homeomorphism, the map $v \mapsto g(x)(v)$ must be a homeomorphism, and hence $k = n$. Since restriction to each fiber is a bijection, $g(x)$ must be an invertible linear map and hence continuous. Hence $\phi \circ f^{-1} \circ \psi^{-1}$ is continuous. Thus f is continuous. ■

Definition 1.1.9 (Sections)

Let k be a field. Let $p : E \rightarrow B$ be a vector bundle over k . A section of p is a map $s : B \rightarrow E$ such that $p \circ s = \text{id}_B$.

Lemma 1.1.10

Let k be a field. Let $p : E \rightarrow B$ be a vector bundle over k . Then the following are true.

- Let $s_1, s_2 : B \rightarrow E$ be sections of p . Then $s_1 + s_2$ is a section of p .
- Let $s : B \rightarrow E$ be a section of p . Let $\lambda \in k$. Then λs is a section of p .

Definition 1.1.11 (The Space of Sections)

Let F be a field. Let $p : E \rightarrow B$ be a vector bundle over F . Define the F -vector space of sections to be the set

$$s(E) = \{s : B \rightarrow E \mid s \text{ is a section of } p\}$$

together with addition and scalar multiplication defined pointwise.

Lemma 1.1.12

Let F be a field. Let $p : E \rightarrow B$ be a vector bundle over F . Then E is isomorphic to the trivial bundle if and only if there exists global sections $s_1, \dots, s_n \in s(E)$ such that $s_1(b), \dots, s_n(b)$ is a basis for $p^{-1}(b)$ for all $b \in B$.

Example 1.1.13

The Möbius vector bundle $M \rightarrow S^1$ is non-trivial.

Proof

Suppose that it is trivial. By the above lemma, there exists a global section $s : S^1 \rightarrow M$ such that $s(e^{2\pi it}) \neq [(t, 0)]$ for all $t \in [0, 1]$. Writing $M = \frac{I \times \mathbb{R}}{\sim}$ where $(0, x) \sim (1, -x)$, we see that the map s is of the form $s(e^{2\pi it}) = [(t, f(t))]$ for some continuous map $f : I \rightarrow \mathbb{R}$ such that $f(0) = -f(1)$ by definition of the equivalence relation. But by the intermediate value theorem, there exists $c \in I$ such that $f(c) = 0$, contradicting the fact that s is nowhere vanishing. ■

Definition 1.1.14 (Set of Isomorphism Classes of Vector Bundles)

Let X be a space. Let F be a field. Denote the set of isomorphism classes of vector bundles by

$$\text{Vect}_n^F(X) = \frac{\{p : E \rightarrow B \mid E \text{ is a vector bundle of } B \text{ over } F\}}{\sim}$$

where $E \sim E'$ if E and E' are isomorphic bundles.

1.2 The Cocycle Conditions

Given two charts (U_α, ϕ_α) and (U_β, ϕ_β) of a vector bundle,

$$\phi_\beta \circ \phi_\alpha^{-1} : (U_\alpha \cap U_\beta) \times F^k \rightarrow (U_\alpha \cap U_\beta) \times F^k$$

is a well defined function. In particular, by fixing a point in $U_\alpha \cap U_\beta$, we obtain a linear map.

Definition 1.2.1 (Transition Functions)

Let $p : E \rightarrow B$ be an F -vector bundle of rank k . Let (U_α, ϕ_α) and (U_β, ϕ_β) be local trivialization. Define $g_{\alpha, \beta} : U_\alpha \cap U_\beta \rightarrow \text{GL}(n, F)$ by

$$x \mapsto \phi_\beta \circ \phi_\alpha^{-1}(x, -) : F^k \rightarrow F^k$$

In other words, $g_{\alpha, \beta}$ is such that

$$\phi_\beta \circ \phi_\alpha^{-1}(x, v) = (x, g_{\alpha, \beta}(x)v)$$

For $x \in U_\alpha \cap U_\beta$ and $v \in F^k$.

Proposition 1.2.2

Let $p : E \rightarrow B$ be a K -vector bundle of rank k . The transition functions of the vector bundle satisfies the following.

- Cocycle condition: $g_{\gamma, \alpha}(x) \circ g_{\beta, \gamma}(x) \circ g_{\alpha, \beta}(x) = \text{id}$ on $U_\alpha \cap U_\beta \cap U_\gamma$ for all $x \in U_\alpha \cap U_\beta \cap U_\gamma$.
- $g_{\alpha, \alpha}(x) = I_r$ for all $x \in U_\alpha$.

Proof

We have

$$\phi_\alpha \circ \phi_\gamma^{-1} \circ \phi_\gamma \circ \phi_\beta^{-1} \circ \phi_\beta \circ \phi_\alpha^{-1} = \text{id}$$

Hence

$$(x, (g_{\gamma, \alpha}(x) \circ g_{\beta, \gamma}(x) \circ g_{\alpha, \beta}(x))(v)) = (x, v)$$

for all $x \in U_\alpha \cap U_\beta \cap U_\gamma$ and all $v \in F^k$. Hence the cocycle condition is achieved.

Clearly $g_{\alpha, \alpha}(x, -)$ is the identity by definition. ■

The cocycle conditions completely determine a vector bundle.

Definition 1.2.3 (Associated Vector Bundle)

Let F be a field. Let B be a space. Let $\{U_i \mid i \in I\}$ be an open cover of B . Let $g_{i, j} : U_i \cap U_j \rightarrow \text{GL}(k, F)$ be a continuous functions for $i, j \in I$ such that $g_{k, i}(x) \circ g_{j, k}(x) \circ g_{i, j}(x) = \text{id}$ for all

$x \in U_i \cap U_j \cap U_k$ and $g_{i,i}(x) = \text{id}$ for all $x \in U_i$. Define the associated vector bundle to be the space

$$E_g = \frac{\coprod_{i \in I} U_i \times F^k}{\sim}$$

where $(i, b, v) \sim (j, b, g_{i,j}(b)(v))$ if $b \in U_i \cap U_j$, together with the induced projection map $\text{proj} : E_g \rightarrow B$.

Lemma 1.2.4

Let F be a field. Let B be a space. Let $\{U_i \mid i \in I\}$ be an open cover of B . Let $g_{i,j} : U_i \cap U_j \rightarrow \text{GL}(k, F)$ be a continuous functions for $i, j \in I$. Then $p : E_g \rightarrow B$ is a vector bundle of rank k .

Proof

Firstly note that the projection map $E_g \rightarrow B$ is well defined by the universal property of quotient. We then need to prove that the local triviality condition is satisfied. Clearly the composition $U_i \times F^k \hookrightarrow \coprod_{i \in I} U_i \times F^k \rightarrow E_g$ composed with the canonical projection map commutes. It is a homeomorphism following the universal property of quotients. ■

Lemma 1.2.5

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let $g = \{g_{i,j} \mid i, j \in I\}$ be the transition functions. Then the transition functions induce an isomorphism

$$E_g \cong E$$

of vector bundles.

1.3 Operations on Vector Bundles

Definition 1.3.1 (Whitney Sum) Let $p_1 : E_1 \rightarrow B$ and $p_2 : E_2 \rightarrow B$ be two vector bundles. Define the direct sum of the vector bundles to be

$$E_1 \oplus E_2 = \{(v_1, v_2) \in E_1 \times E_2 \mid p_1(v_1) = p_2(v_2)\}$$

together with the projection $p : E_1 \oplus E_2 \rightarrow B$ defined by $(v_1, v_2) \mapsto p_1(v) = p_2(v)$.

Example 1.3.2

Let $p : M \rightarrow S^1$ be the Mobius vector bundle. Then $M \oplus M \cong e^2$.

Proof

Define a morphism $M \oplus M \rightarrow e^2$ by $([(t, x)], [(t, y)]) \mapsto (e^{2\pi it}, x, y)$. Notice that that restricted to the open cover $U_1 = \{e^{2\pi it} \mid 0 \leq t \leq 1/2\}$ and $U_2 = \{e^{2\pi it} \mid 1/2 \leq t \leq 1\}$ of S^1 . we see that the local trivialization charts of $M \oplus M$ coincides with that of our morphism. Hence the morphism is a well defined morphism of vector bundles. Moreover, it is an isomorphism of vector spaces on each fiber. ■

Definition 1.3.3 (Pullback Bundle)

Let F be a field. Let X, Y be spaces. Let $p : E \rightarrow Y$ be a vector bundle over F . Let $f : X \rightarrow Y$ be a map. Define the pullback bundle of f to be the space

$$f^*(E) = \{(x, e) \in X \times E \mid f(x) = p(e)\}$$

together with the projection map $p^* : f^*(E) \rightarrow X$ given by $p^*(x, e) = x$.

Definition 1.3.4 (The Hom Bundle)

Let F be a field. Let B be a space. Let $p_1 : E_1 \rightarrow B$ and $p_2 : E_2 \rightarrow B$ be vector bundles over F .

Define the Hom bundle of E_1 and E_2 to be the space

$$\text{Hom}(E_1, E_2) = \coprod_{b \in B} \text{Hom}_F(p_1^{-1}(b), p_2^{-1}(b))$$

together with the coproduct of the maps $\text{Hom}_F(p_1^{-1}(b), p_2^{-1}(b)) \rightarrow \{b\}$.

Definition 1.3.5 (The Dual Bundle)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Define the dual bundle of p to be the vector bundle

$$E^* = \text{Hom}(E, e^1)$$

Definition 1.3.6 (The Tensor Product Bundle)

Let F be a field. Let B be a space. Let $p_1 : E_1 \rightarrow B$ and $p_2 : E_2 \rightarrow B$ be vector bundles over F . Define the tensor product bundle to be the space

$$E_1 \otimes E_2 = \coprod_{b \in B} p_1^{-1}(b) \otimes_F p_2^{-1}(b)$$

together with the coproduct of the maps $p_1^{-1}(b) \otimes p_2^{-1}(b) \rightarrow \{b\}$.

Definition 1.3.7 (The Exterior Product Bundle)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle. Define the n th exterior product bundle of E to be

$$\Lambda^n(E) = \coprod_{b \in B} \Lambda^n(p^{-1}(b))$$

together with the coproduct of the maps $\Lambda^n(p^{-1}(b)) \rightarrow \{b\}$.

1.4 Homotopy Theory of Vector Bundles

Theorem 1.4.1 (The Homotopy Theorem)

Let X be a paracompact space. Let $p : E \rightarrow X \times I$ be a vector bundle. Then there is a bundle isomorphism

$$p^{-1}(X \times \{0\}) \cong p^{-1}(X \times \{1\})$$

Proposition 1.4.2

Let X, Y be paracompact spaces. Let $p : E \rightarrow Y$ be a vector bundle. Let $f, g : X \rightarrow Y$ be homotopic maps. Then there is an isomorphism

$$f^*(E) \cong g^*(E)$$

Corollary 1.4.3

Let X, Y be paracompact spaces. Let $f : X \rightarrow Y$ be a map. If f is a homotopy equivalence, then the pullback map induces natural bijection

$$f^* : \text{Vect}_n^F(Y) \xrightarrow{1:1} \text{Vect}_n^F(X)$$

Corollary 1.4.4

Let B be a paracompact space. Let $p : E \rightarrow B$ be a vector bundle. If B is contractible, then E is trivial.

1.5 Almost Complex Structures

Definition 1.5.1 (Linear Complex Structure)

Let V be a vector space over $\mathbb{R}$. A linear complex structure on V is a linear map $J : V \rightarrow V$ such that $J^2 = -\text{id}_V$.

Proposition 1.5.2

Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V . Then V is a $\mathbb{C}$ -vector space whose action is given by

$$(x + iy) \cdot v = xv + yJ(v)$$

for $v \in V$ and $x + iy \in \mathbb{C}$.

Proposition 1.5.3

Let V be a finite dimensional vector space over $\mathbb{R}$. Then the following are true.

- V admits a linear complex structure if and only if $\dim_{\mathbb{R}}(V)$ is even.
- In this case, we have $\dim_{\mathbb{C}}(V) = 2 \dim_{\mathbb{R}}(V)$.

Example 1.5.4

Let $J : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ be the map defined in block matrix form

$$J = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$$

Then J defines a linear complex structure on $\mathbb{R}^{2n}$. In particular, if $e_1, \dots, e_{2n}$ are the standard basis vectors for $\mathbb{R}^{2n}$, then $\frac{1}{2}(e_1 - ie_{2n+1}), \dots, \frac{1}{2}(e_n - ie_{2n})$ is a basis for V as a $\mathbb{C}$ -vector space.

Proposition 1.5.5

Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V . Let $T : V \rightarrow V$ be an $\mathbb{R}$ -linear transformation. Then T is a $\mathbb{C}$ -linear transformation if and only if $J \circ T = T \circ J$.

The linear complex structure on an $\mathbb{R}$ -vector space gives an intrinsic way of giving the $\mathbb{R}$ -vector space the structure of a $\mathbb{C}$ -vector space. However, we already know an external way of giving $\mathbb{R}$ -vector spaces $\mathbb{C}$ -multiplication: namely by extension of scalars.

Proposition 1.5.6

Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V . Then the following are true.

- J extends to a linear complex structure on $V \otimes_{\mathbb{R}} \mathbb{C}$ by $J(v \otimes z) = J(v) \otimes z$.
- The eigenvalues of $J : V \otimes_{\mathbb{R}} \mathbb{C} \rightarrow V \otimes_{\mathbb{R}} \mathbb{C}$ is given by $\pm i$ and the eigenspaces of J are given by

$$E_i = \{v \otimes 1 - J(v) \otimes i \mid v \in V\} \quad \text{and} \quad E_{-i} = \{v \otimes 1 + J(v) \otimes i \mid v \in V\}$$

- There is a natural isomorphism of $\mathbb{C}$ -vector space $V \cong E_i$ given by $v \mapsto v \otimes 1 - J(v) \otimes i$.
- The conjugation map $\overline{(-)} : V \otimes_{\mathbb{R}} \mathbb{C} \rightarrow V \otimes_{\mathbb{R}} \mathbb{C}$ given by $\overline{v \otimes z} = v \otimes \bar{z}$ restricts to an isomorphism

$$\overline{(-)}|_{E_i} : E_i \xrightarrow{\cong} E_{-i}$$

whose inverse is given by $\overline{(-)}|_{E_{-i}}$.

Proposition 1.5.7

Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V . Then J^T defines a linear complex structure on V^* .

Definition 1.5.8 (Almost Complex Structure)

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over $\mathbb{R}$. An almost complex structure on p is a

vector bundle isomorphism $j : E \rightarrow E$ such that $j \circ j = \text{id}$.

Proposition 1.5.9

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over $\mathbb{R}$. Let J be an almost complex structure on p . Then the following are true.

- The rank of E over $\mathbb{R}$ is even.
- E is a complex vector bundle whose action is given by

$$(x + iy) \cdot v = xv + yJ_b(v)$$

for $v \in E_b$ and $x + iy \in \mathbb{C}$ for each $b \in B$.

Proposition 1.5.10

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over $\mathbb{C}$ of rank n . Then E is a real vector bundle of rank $2n$.

2 Sequences of Vector Bundles

2.1 Subbundles and Quotient Bundles

Definition 2.1.1 (Vector Subbundles)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . A subbundle of E is a subset $E' \subseteq E$ such that $p|_{E'} : E' \rightarrow B$ is a vector bundle over F .

Lemma 2.1.2

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let E' be a subbundle of E . Let $\phi_i : p^{-1}(U_i) \rightarrow U_i \times F^k$ be a local chart of E' . Then the following are true.

- There exists a map $\psi_i : p^{-1}(U_i) \rightarrow U_i \times F^{n-k}$ such that the map $p^{-1}(U_i) \rightarrow U_i \times F^n$ given by $x \mapsto (x, \text{proj}_{F^k}(\phi_i(x)), \text{proj}_{F^{n-k}}(\psi_i(x)))$ is a local chart of E .
- The transition functions of E are given in block matrix

$$g_{ij} = \begin{pmatrix} g'_{ij} & l_{ij} \\ 0 & k_{ij} \end{pmatrix}$$

for some maps $l_{ij} : U_i \cap U_j \rightarrow M_{k,n-k}(F)$ and $k_{ij} : U_i \cap U_j \rightarrow \text{GL}(n-k, F)$.

- The maps k_{ij} satisfy the cocycle condition and $k_{ii}(x) = I_{n-k}$ for all $x \in U_i$.

Definition 2.1.3 (The Quotient Bundle)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let E' be a subbundle of E . Define the quotient bundle E/E' to be the associated vector bundle to the collection of functions k_{ij} given by the above.

Lemma 2.1.4

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let E' be a subbundle of E . There are linear isomorphisms

$$(E/E')_b \cong E_b/E'_b$$

for all $b \in B$.

Proposition 2.1.5 (Universal Property of Quotient Bundles)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let E' be a subbundle of E . Then the quotient bundle and the quotient map $q : E \rightarrow E/E'$ satisfies the following universal property. For any vector bundle G over F and any morphism $f : E \rightarrow G$ such that $f(E')$ is the trivial bundle, there exists a unique morphism $\tilde{f} : E/E' \rightarrow G$ such that the following diagram commutes:

$$\begin{array}{ccc} E & \xrightarrow{q} & E/E' \\ & \searrow f & \downarrow \tilde{f} \\ & & G \end{array}$$

Moreover, E/E' is the unique (up to unique isomorphism) vector bundle that satisfies the above.

2.2 The Kernel of a Morphism

2.3 Short Exact Sequences of Vector Bundles

Definition 2.3.1 (Short Exact Sequences of Vector Bundles)

Let F be a field. Let B be a space. Let E, G, H be vector bundles over B . Let $f : E \rightarrow G$ and $g : G \rightarrow H$ be morphisms. We say that the sequence

$$0 \longrightarrow E \xrightarrow{f} G \xrightarrow{g} H \longrightarrow 0$$

is a short exact sequence of vector bundles if the induced linear maps

$$0 \longrightarrow E_b \xrightarrow{f|_{E_b}} G_b \xrightarrow{g|_{E_b}} H_b \longrightarrow 0$$

are short exact sequences of vector spaces for all $b \in B$.

Lemma 2.3.2

Let F be a field. Let B be a space. Suppose that the following is a short exact sequence of vector bundles over B :

$$0 \longrightarrow E \xrightarrow{f} G \xrightarrow{g} H \longrightarrow 0$$

Then the following are true.

- g induces an isomorphism $G/E \cong H$.
- f induces an isomorphism $\ker(G \rightarrow H) \cong E$.

TBA: Exactness of taking sections TBA: Exactness of tensor products

Note: category of vector bundles is not an abelian category.

3 Reduction of Structure Group

3.1 Structure Groups

4 Metrics on Vector Bundles

4.1

Definition 4.1.1 (Metrics)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle.

- Suppose that $F = \mathbb{R}$. A metric on E is a smooth section of $s \in (E \otimes E)^* \cong E^* \otimes E^*$ such that $s_p : E_b \times E_b \rightarrow \mathbb{R}$ is an inner product for all $b \in B$.
- Suppose that $F = \mathbb{C}$. A metric on E is a smooth section of $s \in (E \otimes E)^* \cong E^* \otimes E^*$ such that $s_p : E_b \times E_b \rightarrow \mathbb{R}$ is a Hermitian product for all $b \in B$.

Lemma 4.1.2

Let $F = \mathbb{R}$ or $\mathbb{C}$. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle. If B is paracompact, then E admits a metric.

Note: ses of vector bundles of paracompact spaces are split exact. Note: vector bundles have reduction of structure group to $O(n)$ and $U(n)$ when they admit a metric.

5 The Topology of Fiber Bundles

5.1 Fiber Bundles

Fiber bundles serve as somewhat of a generalization of both vector bundles and covering spaces, while being a special case of a fibration. It therefore has the properties of a fibration.

Definition 5.1.1 (Fiber Bundles)

Let E, B, F be spaces with B connected, and $p : E \rightarrow B$ a continuous map. We say that p is a fiber bundle over F if the following are true.

- $p : E \rightarrow B$ is surjective
- Local Triviality: There exists a collection

$$\{(U_k \subseteq B, \phi_k : p^{-1}(U_k) \xrightarrow{\cong} U_k \times F) \mid k \in I\}$$

called local trivialization charts such that the following diagram commutes:

$$\begin{array}{ccc} p^{-1}(U_k) & \xrightarrow{\phi_k} & U_k \times F \\ & \searrow p & \swarrow \pi \\ & U_k & \end{array}$$

where π is the projection by forgetting the second variable.

We say that B is the base space, E the total space. It is denoted as (F, E, B) or $F \hookrightarrow E \rightarrow B$.

Intuitively, we would like a fiber bundle to locally look like the product $B \times F$.

Lemma 5.1.2

Let K be a field. Let $p : E \rightarrow B$ be a fiber bundle with fiber F . Then the local trivialization charts induces a homeomorphism

$$p^{-1}(b) \cong F$$

for all $b \in B$.

Fiber bundles generalize vector bundles in the sense that the fibers are no longer vector spaces but instead arbitrary spaces.

Lemma 5.1.3

The following are true regarding fiber bundles.

- Every vector bundle is a fiber bundle.
- Every covering space is a fiber bundle.

Proof Indeed if $p : E \rightarrow B$ is a vector bundle, then each fiber $p^{-1}(b)$ is an n -dimensional vector spaces over a field F . Moreover, by definition the local triviality condition is also satisfied.

Finally, for any $U \subseteq X$, recall that

$$p^{-1}(U) = \coprod_{i \in I} V_i$$

where each $V_i \cong U$. It is clear by definition that $|p^{-1}(x)| = |I|$ for any $x \in X$. By giving I the discrete topology, we obtain a homeomorphism $p^{-1}(x) \cong I$. The homeomorphism $p^{-1}(U) = \coprod_{i \in I} V_i$ translates to

$$p^{-1}(U) = \coprod_{i \in I} V_i \cong \coprod_{i \in I} U \cong U \times I$$

defined by $\tilde{x} \in V_i \mapsto (p(\tilde{x}) = x, i)$. It is thus clear that the local triviality condition is satisfied. ■

Example 5.1.4

The following maps are fiber bundles.

- Let $f_n : S^1 \rightarrow S^1$ be the map defined by $f_n(z) = z^n$. Then f_n is a fiber bundle with fiber the

n th roots of unity.

- Let $f : \mathbb{R} \rightarrow S^1$ be the map defined by $f(t) = e^{2\pi it}$. Then f is a fiber bundle with fiber $\mathbb{Z}$.
- Let $p : S^n \rightarrow \mathbb{RP}^n$ be the projection map. Then p is a fiber bundle with fiber as a two point set with the discrete topology.

Proof

Indeed, all of these are covering spaces and hence fiber bundles by the above lemma. ■

Example 5.1.5

Let $M = \frac{[-1,1] \times [-1,1]}{(t,-1) \sim (1-t,1)}$ be the Mobius strip. Then the projection map $p : M \rightarrow C$ defined by $p(t, s) = s$ is a fiber bundle with fiber $[-1, 1]$.

Proposition 5.1.6 Every fiber bundle is a Serre fibration.

We can provide a partial converse for the fact that every fiber bundle is a Serre fibration.

Proposition 5.1.7 Let $p : E \rightarrow B$ be a fiber bundle. If B is paracompact, then p is a (Hurewicz) fibration.

We there fore have inclusions

$$\text{Fiber Bundles} \subset \text{Serre Fibrations} \subset \text{(Hurewicz) Fibrations}$$

Definition 5.1.8 (Map of Fiber Bundles)

Let (F_1, E_1, B_1) and (F_2, E_2, B_2) be fiber bundles. A map of fiber bundles is a pair of basepoint preserving continuous maps $(\tilde{f} : E_1 \rightarrow E_2, f : B_1 \rightarrow B_2)$ such that the following diagram commutes:

$$\begin{array}{ccc} E_1 & \xrightarrow{\tilde{f}} & E_2 \\ p_1 \downarrow & & \downarrow p_2 \\ B_1 & \xrightarrow{f} & B_2 \end{array}$$

A map of fiber bundles $(\tilde{f}, f)$ is said to be an isomorphism if there is a map $(\tilde{g} : E_2 \rightarrow E_1, g : B_2 \rightarrow B_1)$ such that $\tilde{g}$ is the inverse of $\tilde{f}$ and g is the inverse of f .

Such a map of fiber bundles determine a continuous of the fibers $F_1 \cong p_1^{-1}(b_1) \rightarrow p_2^{-1}(b_2) \cong F_2$. Moreover, such a morphism preserves fibers. Indeed, If $p_1^{-1}(b)$ is a fiber of B , then using the commutativity of the diagram we have that

$$p_2(\tilde{f}(p_1^{-1}(b))) = f(p_1(p_1^{-1}(b))) = f(b)$$

which implies that

$$p_2^{-1}(f(b)) = \tilde{f}(p_1^{-1}(b))$$

or in other words, the fiber at $f(b)$ is the same as the fiber at b applied with $\tilde{f}$.

Definition 5.1.9 (Isomorphic Fiber Bundles)

Let $p_1 : E_1 \rightarrow B_1$ and $p_2 : E_2 \rightarrow B_2$ be two fiber bundles. Let $(\tilde{f}, f) : p_1 \rightarrow p_2$ be a map of fiber bundles. We say that $(\tilde{f}, f)$ is an isomorphism if there exists a map $(\tilde{g}, g) : p_2 \rightarrow p_1$ of fiber bundles such that $\tilde{f}$ is the inverse of $\tilde{g}$ and f is the inverse of g . In this case, we say that p_1 and p_2 are isomorphic.

There are two important special cases of fiber bundles that will appear time and time again.

Definition 5.1.10 (Trivial Bundles)

We say that a fiber bundle (F, E, B) is trivial if (F, E, B) is isomorphic to the trivial fibration $B \times F \rightarrow B$.

Definition 5.1.11 (The Pullback Bundle) Let $p : E \rightarrow B$ be a fiber bundle with fiber F . Let $f : B' \rightarrow B$ be a continuous function. Define the pullback of p by f to be the space

$$f^*(E) = \{(b', e) \in B' \times E \mid p(e) = f(b')\}$$

Theorem 5.1.12 Let $p : E \rightarrow B$ be a fiber bundle. Suppose that $f, g : X \rightarrow B$ are homotopic maps. Then the pull back bundles

$$f^*(E) \cong g^*(E)$$

are equivalent.

5.2 Sections of a Bundle

Definition 5.2.1 (Sections) Let (F, E, B) be a fiber bundle. A section on the fiber bundle is a map $s : B \rightarrow E$ such that

$$p \circ s = \text{id}_B$$

Definition 5.2.2 (Local Sections) Let (F, E, B) be a fiber bundle. Let $U \subset B$ be an open set. A local section of the fiber bundle on U is a map $s : U \rightarrow E$ such that

$$p \circ s = \text{id}_U$$

5.3 Sphere Bundles

We now consider a special type of fibrations where the fibers are given by S^1 . We define a class of fiber bundles of the form $S^1 \rightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$. When $n = 1$, this is the Hopf fibration.

Definition 5.3.1 (Sphere Bundles) A sphere bundle is a fiber bundle $p : E \rightarrow B$ for which its fibers are the n -sphere S^n .

Example 5.3.2

Let $n \in \mathbb{N}$. Consider S^{2n+1} lying inside $\mathbb{C}^{n+1}$. The projection map $\mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{CP}^n$ induces a fiber bundle

$$S^1 \hookrightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$$

Proof

Clearly the projection map $p : S^{2n+1} \rightarrow \mathbb{CP}^n$ is surjective. Consider the open cover $\{U_i \mid 0 \leq i \leq n\}$ where $U_i = \{[z_0 : \dots : z_n] \mid z_i \neq 0\}$. Define the local trivialization charts $\phi_i : p^{-1}(U_i) \rightarrow U_i \times S^1$ by $(z_0, \dots, z_n) \mapsto ([z_0 : \dots : z_n], z_i/|z_i|)$. It is continuous since it is the product of the composition map and the projection map to the sphere S^1 . Its inverse is the continuous map $([z_0, \dots, z_n], w) = w|z_i|z_i^{-1}(z_0, \dots, z_n)$ since it is induced by the universal property of quotients. It is clear that the required diagram commutes, that is $p = \text{proj} \circ \phi_i$ where $\text{proj} : U_i \times S^1 \rightarrow U_i$ is the projection map to the first coordinate. ■

Example 5.3.3

For each $n \in \mathbb{N}$, we have a map of fiber bundles:

$$\begin{array}{ccc} S^{2n+1} & \hookrightarrow & S^{2n+3} \\ \downarrow & & \downarrow \\ \mathbb{CP}^n & \hookrightarrow & \mathbb{CP}^{n+1} \end{array}$$

whose horizontal maps are induced by the inclusion $\mathbb{C}^n \hookrightarrow \mathbb{C}^{n+1}$ into the first n coordinates. Taking colimit of the diagram, we have that

$$S^1 \hookrightarrow S^\infty \rightarrow \mathbb{CP}^\infty$$

is a fiber bundle.

Proof

Since the local triviality condition is a local condition, for each $x \in \mathbb{CP}^\infty$, x lives in $\mathbb{CP}^n$ for some $n \in \mathbb{N}$ and since $S^1 \hookrightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$ is a fiber bundle, there exists a chart for which the local triviality condition is satisfied. ■

Example 5.3.4 (Hopf Fibration / Hopf Bundle)

Consider the above example with $n = 1$. Restricting the total space to S^3 , the fiber bundle $p : S^3 \rightarrow S^2$ with fiber S^1 is called the Hopf fibration / Hopf bundle.